Session 9: The spike

How to use this booklet. Each Spike Lab part has its own page, with the time, who to work with, and the steps. You may always work alone. **Lost during class?** Page 2 has the whole session on one page; every term is on page 3.

Part	What we do	Time
Opening	One language for every sense; open Spike Lab	12 min
1 The spike	Threshold and all-or-none (Lab Part 1, page 4)	18 min
2 The doors	Sodium up, potassium down (Lab Part 2, page 5)	22 min
3 The pause	The refractory period (Lab Part 3, page 6)	13 min
4 The code	Adrian's discoveries, rate coding (Lab Part 4, page 7)	17 min
Wrap-up	Today on one page, again; exit ticket	8 min

Warm-up

What travels from your eyes, ears and nose to your brain? The same thing for all three, or different? My guess:

_____

Exit ticket (end of class, 3 minutes)

1. What does "all or none" mean?

   _____

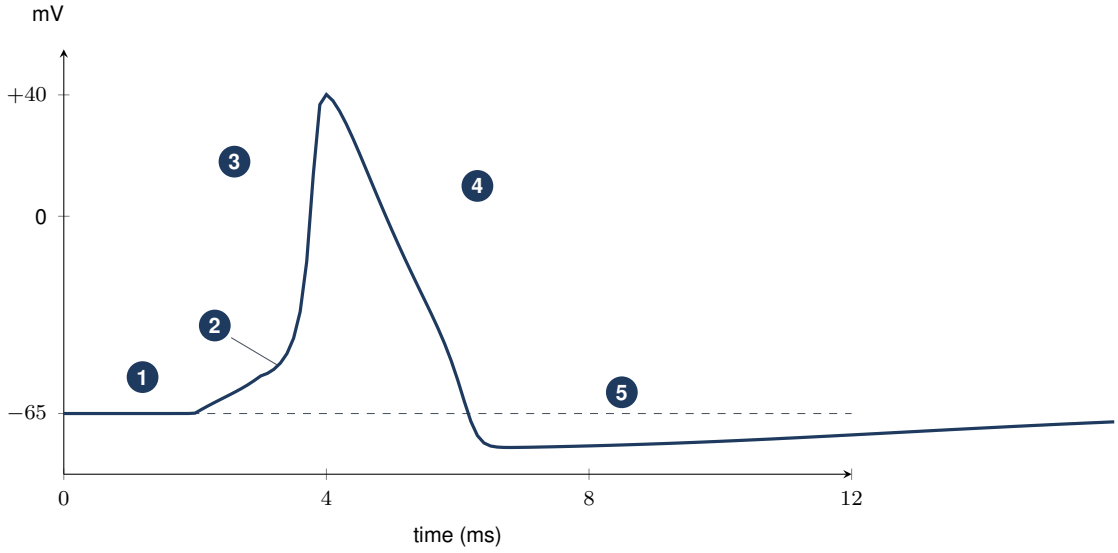
2. Which ion brings the voltage up? _____ Which brings it down? _____

3. How does a neuron signal a stronger stimulus?

   _____

Today on one page

One spike, five moments. If you lose the thread, find the number here.



1 **Rest.** The inside sits at about -65 mV. **Na**$^+$ is crowded outside, **K**$^+$ inside.

2 **Threshold.** A push makes the inside less negative. Past threshold, the spike happens; below it, nothing.

3 **Up.** Voltage-gated **Na**$^+$ doors open; **Na**$^+$ rushes in; more doors open (a chain reaction). The inside reaches about $+40$ mV.

4 **Down.** **Na**$^+$ doors lock shut; slower **K**$^+$ doors open; **K**$^+$ flows out and the inside turns negative again.

5 **Dip and pause.** **K**$^+$ doors close slowly, so the voltage dips below rest. For a short time the neuron cannot fire again.

Spike Lab (runs in any browser; nothing is saved or sent)

Open it on a laptop or phone and keep it beside the slides. Each slide marked **SPIKE LAB** says which part to do.

rakesh-sengupta.github.io/neural-signalling-labs/labs/spike-lab

Word list

Keep this beside you during the lab.

spike / **action potential**	a brief, all-or-none flip of a neuron's voltage, lasting about 2 ms
resting potential	the voltage inside a neuron at rest: about -65 mV
threshold	the voltage a push must reach for a spike to happen (often around -55 mV)
all or none	a spike either happens at full size or not at all
depolarise	make the inside less negative (towards 0)
voltage-gated channel	a protein door that opens or closes when the voltage changes
inactivation	sodium doors locking shut by themselves, about 1 ms after opening
rising phase / **overshoot**	the fast climb of the spike / the part above 0 mV
falling phase / **undershoot**	the return / the dip below rest
refractory period	the pause after a spike: *absolute* (cannot fire) then *relative* (needs a bigger push)
TTX	a puffer-fish toxin that blocks sodium doors
TEA	a chemical that blocks potassium doors
rate coding	signalling stimulus strength by the number of spikes per second
adaptation	the firing rate falling during a constant stimulus
spike train	a sequence of spikes over time
homunculus	the imagined "little person" in the head who watches the senses

Spike Lab, Part 1: find the threshold

Time: 8 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

☐ Open Spike Lab (page 2) and press **Start Part 1**.

☐ Set a strength and press **Stimulate**. Each press is one short push (1 ms).

☐ Find the **weakest** strength that makes a spike.

☐ Read the peak at that strength, then at strength 10. Type your numbers into the lab and press **Check**.

Predict first. If you double the push, the spike will be: ☐ twice as tall ☐ about the same ☐ smaller

Weakest strength that makes a spike _____

Peak at that strength _____ mV

Peak at strength 10 _____ mV

In your own words: what does “all or none” mean, using your two peaks?

Spike Lab, Part 2: watch the doors, then block them

Time: 8 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

☐ Press **Stimulate** and watch the lower panel (**orange**: sodium doors; **purple**: potassium doors).

☐ Tick the **sodium blocker** (like TTX) and stimulate.

☐ Untick it, tick the **potassium blocker** (like TEA) and stimulate.

☐ Answer the three questions in the lab.

Which doors open first? ☐ sodium ☐ potassium

With the sodium blocker, the spike. . . _____

Time above 0 mV, no blockers _____ ms

Time above 0 mV, potassium blocker _____ ms

Finish the sentences. The voltage goes **up** because _____ .

It comes back **down** because _____ .

Link to Session 7. Why does the dentist's lidocaine stop you feeling pain?

_____

_____

Spike Lab, Part 3: can it fire again straight away?

Time: 6 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

- ☐ Two pushes now, with a gap. Start at strength 8 and a short gap.
- ☐ Make the gap longer, one step at a time, until **both** pushes make a spike.
- ☐ Set strength 10 and find the shortest gap again.

Shortest gap for two spikes, strength 8 _____ ms

Shortest gap for two spikes, strength 10 _____ ms

1. A stronger push fired sooner. Which refractory period does that show? ☐ absolute ☐ relative

2. The model is a squid axon in cold sea water. In your warm neurons, the absolute refractory period is about 1 ms. What is the most spikes per second a neuron could then fire? (1 second = 1000 ms)

3. Why does the pause make spikes travel in one direction only? (Hint: the patch behind a spike has just fired.)

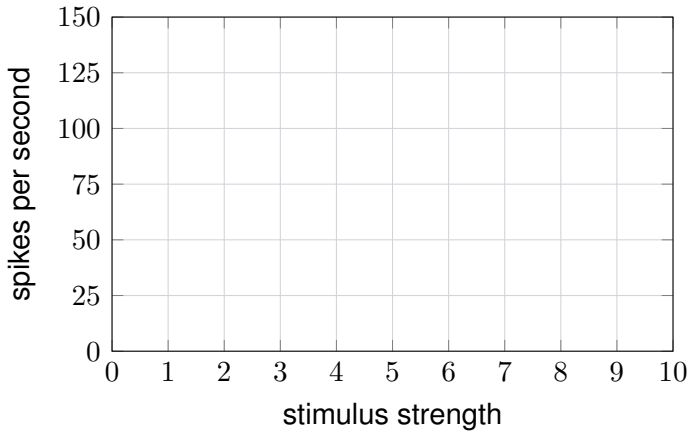
Spike Lab, Part 4: stronger means faster

Time: 6 minutes | **Who:** alone or in pairs | **You need:** a laptop or phone

Steps (tick each one when done)

- ☐ The push now stays on for 200 ms, like a weight hanging on a muscle.
- ☐ Try strengths 1, 3, 5, 7 and 10. Write the spikes per second in the table.
- ☐ Plot your five points on the grid.

Strength	Spikes per second
1	
3	
5	
7	
10	



1. Describe the pattern in one sentence. This is what Edgar Adrian found in 1926.

2. Adrian also found **adaptation**: with a constant stimulus, the rate slowly falls. Give one everyday example of no longer noticing something that is still there.

Study cards

Edgar Adrian and the first single-neuron recordings (1926)

Found: with new valve amplifiers, Adrian recorded single sensory nerve fibres. He showed that their spikes are all or none, that the firing rate rises with stimulus strength (a heavier weight on a frog's muscle gave more spikes per second), and that the rate falls during a constant stimulus (adaptation). Nobel Prize, 1932, shared with Charles Sherrington *(Rieke et al., 1997, Ch. 1)*.

Hodgkin and Huxley: the spike in equations (1952)

Found: using the squid giant axon, they measured sodium and potassium currents separately and wrote equations for doors that open and close with voltage. The equations predicted a pulse with the shape and speed of a real spike. Nobel Prize, 1963. Spike Lab runs these equations.

Neher and Sakmann: one door at a time (1976)

Found: with the patch-clamp method, they recorded the tiny current through a single channel protein as it flicked open and shut. Nobel Prize, 1991.

Why spikes? (Hodgkin and Rushton, 1946)

Found: without active doors, a voltage change spreads and fades within about a millimetre. A spike is rebuilt along the axon, so it can travel a metre, from fingertip to spinal cord, at full size.

Timing matters too (Mainen and Sejnowski, 1995)

Found: a cortical neuron given the same flickering input twice fires at nearly the same moments, to within about a millisecond. Spikes can carry information in *when* they happen, not just how many.

The homunculus (Rieke et al., 1997)

Idea: the brain receives only spike trains. A "little person" watching them explains nothing. Yet when scientists record spikes and ask what they mean, they take that little person's place: understanding the neural code means being able to read the spikes.

Answers (check after class)

Part 1. Threshold at strength **4.5** (at 4.0 the voltage only reaches about -59 mV). Peaks: about $+37$ mV at threshold, about $+41$ mV at strength 10. More than twice the push, nearly the same spike: **all or none**. Prediction: about the same.

Part 2. Sodium doors open first. Sodium blocker: **no spike**. Time above 0 mV: about **1.2 ms** normally, about **2.6 ms** with the potassium blocker. Up: **Na^+** rushes in through voltage-gated doors. Down: **Na^+** doors lock shut and **K^+** flows out. Lidocaine blocks sodium doors in the nerves near the injection, so pain spikes cannot be made or passed on.

Part 3. Shortest gaps: **13 ms** at strength 8, **12 ms** at strength 10. (1) Relative. (2) About $1000 \div 1 =$ **1000 spikes per second** at most; most neurons fire far more slowly. (3) The patch just behind a travelling spike is still refractory, so the spike can only move forwards.

Part 4. About 5 (one spike only), 80, 90, 105 and 115 spikes per second for strengths 1, 3, 5, 7 and 10. (1) The stronger the stimulus, the more spikes per second: **rate coding**. The squid model jumps from one spike to about 65 per second; real sensory neurons can also fire slowly. (2) Examples: a watch on your wrist, clothes on your skin, the hum of a fan, a smell in a room after a while.

Exit ticket. (1) Past threshold a spike is full size; below it, there is no spike. (2) Sodium up, potassium down. (3) By firing more spikes per second (and sometimes by when spikes arrive).

Sources

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- Spike Lab: the Hodgkin–Huxley (1952) model of the squid giant axon at 6.3 °C. The “potassium blocker” removes 90% of the voltage-opened potassium current and keeps the resting part.